THE PUNCTURED SPECTRUM OF A NOETHERIAN LOCAL RING IS AFFINE IF AND ONLY IF THE DIMENSION IS AT MOST ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-affine-punctured-spec is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 3348-3353 and proof span 3355-3407 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The author announces in the proof that the argument given is 'somewhat nonstandard' and that the standard argument is discussed in a remark below; that sentence is kept verbatim, and the remark it refers to is outside the counted span and is not part of this card.

Editorial note. The proof writes the snake lemma sequence and the two length sequences with the author's own notation (Q[f], l_n); all displays were copied unchanged.

Editorial note. 6 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A Noetherian ring has finitely many minimal primes, and an element of the maximal ideal avoiding them can be chosen; the spectrum of a localisation is the corresponding open subset. Cited in the same chapter's algebra parent at proof lines 3358-3361 and reused at 3377-3379. Produces the element f that makes the punctured spectrum a principal affine open in dimension one, and the nonzerodivisor used in the converse.
- Dimension drops by one on cutting out a nonzerodivisor, and a finite ring map induces a closed map on spectra. Cited at proof lines 3382-3383 and again at 3398-3400. Gives the dimension count for A modulo fA and turns the finiteness of the ring map A/fA to B/fB into closedness of the induced map on spectra, which produces the contradiction.
- A module whose support is the closed point of a Noetherian local ring has finite length. Cited at proof lines 3390-3391. Makes the length l of the

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module of f-torsion in Q finite, which is what bounds the lengths of the modules Q[f^n].

Source reference. Original reference algebra-lemma-Noetherian-irreducible-components: sources/stacks-divisors/pinned-original-source.tex, source lines 3360–3360. Same-chapter reference used in the retained author text at source line(s) 3360. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference algebra-lemma-going-up-closed: sources/stacks-divisors/pinned-original-source.tex, source lines 3403–3403. Same-chapter reference used in the retained author text at source line(s) 3403. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference algebra-lemma-integral-going-up: sources/stacks-divisors/pinned-original-source.tex, source lines 3402–3402. Same-chapter reference used in the retained author text at source line(s) 3402. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference algebra-lemma-one-equation: sources/stacks-divisors/pinned-original-source.tex, source lines 3036–3036. Same-chapter reference used in the retained author text at source line(s) 3036, 3378, 5578. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference algebra-lemma-silly: sources/stacks-divisors/pinned-original-source.tex, source lines 3361–3361. Same-chapter reference used in the retained author text at source line(s) 3361, 6306. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference algebra-lemma-support-point: sources/stacks-divisors/pinned-original-source.tex, source lines 3387–3387. Same-chapter reference used in the retained author text at source line(s) 3387. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a Noetherian local ring. The punctured spectrum $U = \mathrm{Spec}(A) \setminus \{\mathfrak{m}\}$ of A is affine if and only if $\dim(A) \leq 1$.*

Proof. If $\dim(A) = 0$, then U is empty hence affine (equal to the spectrum of the 0 ring). If $\dim(A) = 1$, then we can choose an element $f \in \mathfrak{m}$ not contained in any of the finite number of minimal primes of A (Algebra, Lemmas algebra-lemma-Noetherian-irreducible-components and algebra-lemma-silly). Then $U = \mathrm{Spec}(A_f)$ is affine.

The converse is more interesting. We will give a somewhat nonstandard proof and discuss the standard argument in a remark below. Assume $U = \mathrm{Spec}(B)$ is affine. Since affineness and dimension are not affected by going to the reduction we may

replace A by the quotient by its ideal of nilpotent elements and assume A is reduced. Set $Q = B/A$ viewed as an A-module. The support of Q is $\{\mathfrak{m}\}$ as $A_\mathfrak{p} = B_\mathfrak{p}$ for all nonmaximal primes $\mathfrak{p}$ of A. We may assume $\dim(A) \geq 1$, hence as above we can pick $f \in \mathfrak{m}$ not contained in any of the minimal ideals of A. Since A is reduced this implies that f is a nonzerodivisor in A. The same argument shows that $f : B \to B$ is injective as $\mathrm{Spec}(B)$ is open in $\mathrm{Spec}(A)$. Note that $\dim(A/fA) = \dim(A) - 1$ by Algebra, Lemma algebra-lemma-one-equation. Applying the snake lemma to multiplication by f on the short exact sequence $0 \to A \to B \to Q \to 0$ we obtain

$$0 \to Q[f] \to A/fA \to B/fB \to Q/fQ \to 0$$

where $Q[f] = \mathrm{Ker}(f : Q \to Q)$. This implies that $Q[f]$ is a finite A-module. Since the support of $Q[f]$ is $\{\mathfrak{m}\}$ we see $l = \mathrm{length}_A(Q[f]) < \infty$ (Algebra, Lemma algebra-lemma-support-point). Set $l_n = \mathrm{length}_A(Q[f^n])$. The exact sequence

$$0 \to Q[f^n] \to Q[f^{n+1}] \xrightarrow{f^n} Q[f]$$

shows inductively that $l_n < \infty$ and that $l_n \leq l_{n+1}$. Considering the exact sequence

$$0 \to Q[f] \to Q[f^{n+1}] \xrightarrow{f} Q[f^n] \to Q/fQ$$

and we see that the image of $Q[f^n]$ in Q/fQ has length $l_n - l_{n+1} + l \leq l$. Since $Q = \bigcup Q[f^n]$ we find that the length of Q/fQ is at most l, i.e., bounded. Thus Q/fQ is a finite A-module. Hence $A/fA \to B/fB$ is a finite ring map, in particular induces a closed map on spectra (Algebra, Lemmas algebra-lemma-integral-going-up and algebra-lemma-going-up-closed). On the other hand $\mathrm{Spec}(B/fB)$ is the punctured spectrum of $\mathrm{Spec}(A/fA)$. This is a contradiction unless $\mathrm{Spec}(B/fB) = \emptyset$ which means that $\dim(A/fA) = 0$ as desired. $\square$